

Isogonality

Canada IMO Camp

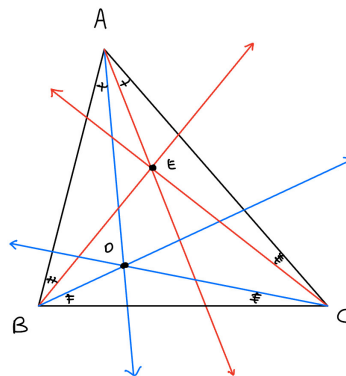
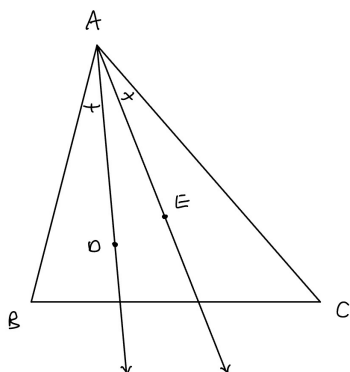
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1 Introduction

Consider a triangle ABC . Given two points D, E , neither of which is one of $\{A, B, C\}$, we say that AD and AE are *isogonal* with respect to $\angle BAC$ iff the two lines are reflections across the angle bisector of $\angle BAC$. We say that D and E are *isogonal conjugates* if:

1. AD and AE are isogonal with respect to $\angle BAC$,
2. BD and BE are isogonal with respect to $\angle CBA$, and
3. CD and CE are isogonal with respect to $\angle ACB$.

Today we'll study a bunch of properties about isogonal lines and isogonal conjugates. We'll even discuss a bit how isogonal conjugates work in *quadrilaterals*.



2 Isogonal Lines

As a starter, let's discuss some properties about isogonal lines. You may have seen isogonal lines already in your geometry studies. For example, in a triangle ABC ,

1. if I is the incenter or *any* excenter, then AI is isogonal to itself with respect to $\angle ABC$;
2. if H is the orthocenter and O the circumcenter, then AH and AO are isogonal with respect to $\angle ABC$.

One can prove these statements with angle chasing. Now, isogonal lines satisfy a lot of useful configurations, such as the following:

Example 2.1. Consider a triangle ABC , and points $D, E \in BC$. Then (ABC) and (ADE) are tangent (at A) iff AD and AE are isogonal with respect to $\angle BAC$.

1. For one proof, let $AD \cap (ABC) = X$, $AE \cap (ABC) = Y$. Prove that XY is parallel to BC iff AD and AE are isogonal.

2. Finish using a homothety.

3. Alternatively, let O be the circumcenter of ABC , and T the circumcenter of ADE . Angle chase to show that $\angle BAO = \angle BAT$ iff AD and AE are isogonal.

Example 2.2. Consider a triangle ABC , and points $D \in BC$ and $E \in (ABC)$. If AD and AE are isogonal with respect to $\angle BAC$, then ABD and AEC are *spirally similar*.

For a proof, consider angle chasing. We can use this lemma to add another pair of isogonal lines to our collection.

Example 2.3. Consider a triangle ABC . Let M be the midpoint of BC , and let the tangents to (ABC) from B and C intersect at K . Then AK and AM are isogonal with respect to $\angle BAC$.

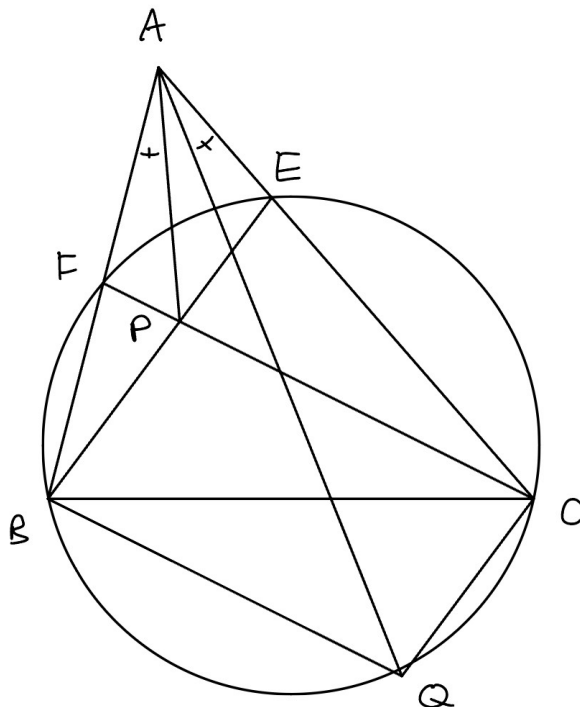
1. First, let $AK \cap (ABC) = T$. Prove using similar triangles that

$$\frac{AB}{AC} = \frac{TB}{TC}.$$

2. Now, consider $N \in BC$ such that AN and AT are isogonal with respect to $\angle BAC$. Using the spiral similarity $ABT \sim ANC$, prove that $NB = NC$. Conclude.

Now, here's a lemma, aptly named the *Isogonal Line Lemma*, which produces a pair of isogonal lines from a common configuration.

Lemma 2.4 (Isogonal Line Lemma). *Consider a triangle ABC and points $E \in AC$, $F \in AB$ such that $BFEC$ is cyclic. Suppose $P = BE \cap CF$, and consider Q such that $BPCQ$ is a parallelogram. Then AP and AQ are isogonal with respect to $\angle BAC$.*



For a proof, note that we may stitch the two similarities $ABC \sim AEF$ and $QBC \sim PCB \sim PEF$ together to get that $ABQC \sim AEFF$, and hence the result.

Note that using this lemma we may show that if H is the orthocenter and O the circumcenter of a triangle ABC then AH and AO are isogonal with respect to $\angle BAC$. Here's a more involved application.

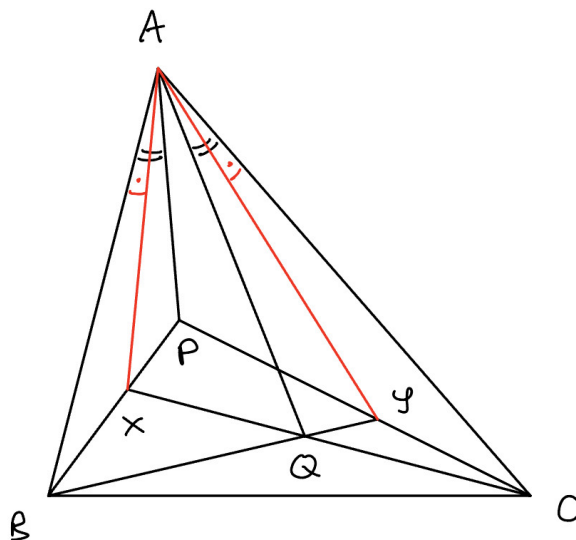
Example 2.5. Let $ABCD$ be a cyclic quadrilateral whose diagonals AC and BD meet at E . The extensions of the sides AD and BC beyond A and B meet at F . Let G be the point such that $ECGD$ is a parallelogram, and let H be the image of E under reflection in AD . Prove that D, H, F, G are concyclic.

As a hint, consider the statement $DHF \sim GCF$.

Here's one more lemma which also shows up sometimes.

Lemma 2.6. (Second Isogonal Line Lemma) Let ABC be a triangle, and consider points P, Q such that AP, AQ are isogonal with respect to $\angle BAC$. Let $X = BP \cap CQ$ and $Y = BQ \cap CP$. Then AX and AY are also isogonal with respect to $\angle BAC$.

We'll leave the proof of this lemma to you.



For more information on the two Isogonal Line Lemmas, see the two blog posts [here](#) and [here](#).

2.1 Exercises

1. Consider a triangle ABC with circumcenter O . Consider a pair of lines l and l' isogonal with respect to $\angle BAC$, and suppose l and l' intersect the perpendicular bisector of BC at P, Q , respectively. Show that $OP \cdot OQ = OA^2$.
2. Let ABC be an acute triangle with $AB < AC$, and let D and E be points on side BC such that $BD = CE$ and D lies between B and E . Suppose there exists a point P inside ABC such that $PD \parallel AE$ and $\angle PAB = \angle EAC$. Prove that $\angle PBA = \angle PCA$.
3. The diagonals of a trapezoid $ABCD$ intersect at point P . Point Q lies between the parallel lines BC and AD such that $\angle AQD = \angle CQB$, and line CD separates points P and Q . Prove that $\angle BQP = \angle DAQ$.
4. Let P be a point inside triangle ABC , and suppose lines AP, BP, CP meet the circumcircle again at T, S, R (here $T \neq A, S \neq B, R \neq C$). Let U be any point in the interior of PT . A line through U parallel to AB meets CR at W , and the line through U parallel to AC meets BS again at V . Finally, the line through B parallel to CP and the line through C parallel to BP intersect at point Q . Given that RS and VW are parallel, prove that $\angle CAP = \angle BAQ$.
5. In cyclic quadrilateral $ABCD$, diagonals AC and BD intersect at P . Let E and F be the respective feet of the perpendiculars from P to lines AB and CD . Segments BF and CE meet at Q . Prove that lines PQ and EF are perpendicular to each other.

6. Let $ABCD$ be a cyclic quadrilateral, and let diagonals AC and BD intersect at X . Let C_1, D_1 and M be the midpoints of segments CX, DX and CD , respectively. Lines AD_1 and BC_1 intersect at Y , and line MY intersects diagonals AC and BD at different points E and F , respectively. Prove that line XY is tangent to the circle through E, F and X .
7. Let E and F be the intersections of opposite sides of a convex quadrilateral $ABCD$. The two diagonals meet at P . Let O be the foot of the perpendicular from O to EF . Show that $\angle BOC = \angle AOD$.
8. Prove the second isogonal line lemma.
9. Let ABC be a scalene triangle, and let D be a point on side BC satisfying $\angle BAD = \angle DAC$. Suppose that X and Y are points inside ABC such that triangles ABX and ACY are similar and quadrilaterals $ACDX$ and $ABDY$ are cyclic. Let lines BX and CY meet at S and lines BY and CX meet at T . Prove that lines DS and AT are parallel.
10. AH is the altitude of triangle ABC and H' is the reflection of H through the midpoint of BC . If the tangent lines to the circumcircle of ABC at B and C , intersect each other at X and the perpendicular line to XH' at H' , intersects AB and AC at Y and Z respectively, prove that $\angle ZXC = \angle YXB$.

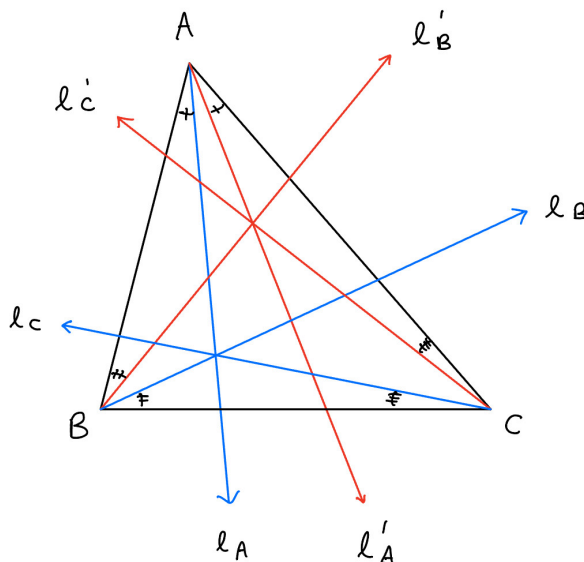
3 Isogonal Conjugates

Recall that given a triangle ABC and points D and E , we call D and E *isogonal conjugates* with respect to ABC if PD and PE are isogonal for $P \in \{A, B, C\}$. Here are some examples of isogonal conjugates:

1. the incenter, and any excenter, are all isogonal conjugates of themselves;
2. the orthocenter and circumcenter are isogonal conjugates of each other;
3. given ABC , define l_A as the line tangent to (ABC) at A , and similarly define l_B and l_C . Let $D = l_B \cap l_C$, $E = l_C \cap l_A$, and $F = l_A \cap l_B$. Then AD, BE, CF concur at the *symmedian point* K , and the isogonal conjugate of K with respect to ABC is the centroid G .

In that last example, it was a rather nice coincidence that AD, BE, CF were concurrent. Or was it? It in fact is true that any point has an isogonal conjugate with respect to any given triangle (which, by our previous results, implies that AD, BE, CF must have been concurrent in the previous example).

Theorem 3.1. Consider a triangle ABC and a point P , which is not one of $\{A, B, C\}$. Then P has an isogonal conjugate with respect to ABC .



Proof. Recall *Ceva's Theorem*, which states that for any lines l_A, l_B, l_C , where l_P passes through P for $P \in \{A, B, C\}$, we have that

$$l_A, l_B, l_C \text{ concur} \iff \frac{\sin \angle(l_A, AB)}{\sin \angle(l_A, AC)} \cdot \frac{\sin \angle(l_B, BC)}{\sin \angle(l_B, BA)} \cdot \frac{\sin \angle(l_C, CA)}{\sin \angle(l_C, CB)} = 1.$$

If l'_A is the isogonal of l_A with respect to $\angle BAC$, then

$$\frac{\sin \angle(l_A, AB)}{\sin \angle(l_A, AC)} = \frac{\sin \angle(l'_A, AC)}{\sin \angle(l'_A, AB)}.$$

Thus l_A, l_B, l_C concur iff l'_A, l'_B, l'_C concur, implying that P has an isogonal conjugate so long as AP, BP, CP are well-defined lines, which is as given. $\square$

This isogonal conjugation correspondence is one-to-one only for the points such that both itself and its isogonal conjugate are not one of $\{A, B, C\}$. These are precisely the points not lying on AB, BC , or CA , so for any such P not on AB, BC, CA , it has a unique isogonal conjugate Q whose unique isogonal conjugate is P .

We'll get into some of the theory of isogonal conjugates soon. But even the *existence* of isogonal conjugates can be a very useful fact!

Example 3.2. (Pascal's Theorem) Given six points A, B, C, D, E, F on a circle, show that the three points $X = AB \cap DE, Y = BC \cap EF, Z = CD \cap FA$ are collinear.

1. Consider the point Z' such that $XAZD$ and $XEZ'B$ are similar figures. What are Y and Z' with respect to XEB ?
2. Using this, show $\angle EXY = \angle ZXD$, and hence prove the theorem.

This theorem is extremely useful, so if you're interested in further discussion of it go see Chapter 9 of Evan Chen's *EGMO*.

Remark 3.3. Why can adding isogonal conjugates be so incredibly powerful? In some sense, when proving the existence of isogonal conjugates, we use an *algebraic* identity (Ceva) to prove a *synthetic* fact. So we're implicitly doing a trig bash, which is why it can feel unusually powerful for a synthetic tool.

3.1 Isogonal Conjugate Theory

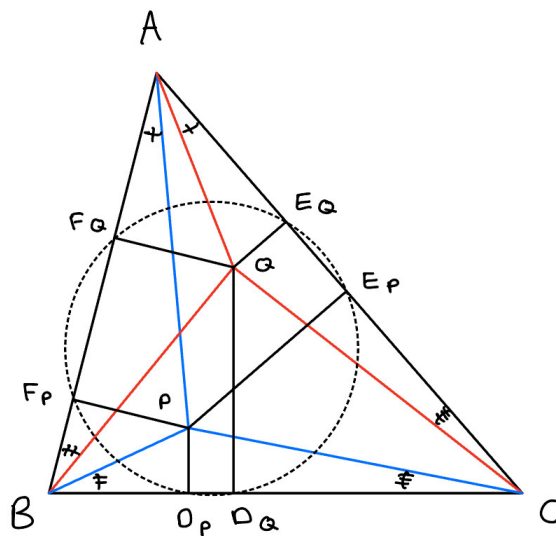
Here are some cool theorems about isogonal conjugates. The first bit of theory we will discuss regards *Pedal circles*.

Definition 3.4. Given a triangle ABC and a point P , let the feet of the altitudes from P to BC, CA, AB be D, E, F respectively. Then the *pedal circle* of P is the circumcircle of DEF .

As a quick check: what are the pedal circles of the incenter, circumcenter, and orthocenter?

Theorem 3.5. Consider a triangle ABC , as well as points P, Q which are isogonal conjugates with respect to ABC . Then P and Q share the same pedal circle.

1. Let the feet of the altitudes from P and Q to BC, CA, AB be $D_P, D_Q, E_P, E_Q, F_P, F_Q$, respectively. Prove that $\angle E_P F_P A = \angle F_Q E_Q A$.
2. Prove $D_P D_Q E_P E_Q, E_P E_Q F_P F_Q, F_P F_Q D_P D_Q$ are all cyclic.
3. Using radical axis, show $D_P D_Q E_P E_Q F_P F_Q$ cyclic.



Using the ideas above one may readily show that if distinct points P and Q share a pedal circle, then P and Q are isogonal conjugates.

Here's an extension of this fact. Before we state it, here is a useful property of ellipses, which we will not prove.

Lemma 3.6. *Consider an ellipse $\mathcal{E}$ with foci A and B , and consider a point $P \in \mathcal{E}$ not on AB . Then the tangent to $\mathcal{E}$ at P externally bisects $\angle APB$.*

As a reminder, the foci A, B of $\mathcal{E}$ are the unique pair of points which satisfy that for all $P \in \mathcal{E}$, $AP + PB$ is always the same length. Here's now a nice property about isogonal conjugates:

Theorem 3.7. *Consider a triangle ABC , and an ellipse $\mathcal{E}$ inscribed within it. Then the foci P and Q of $\mathcal{E}$ are isogonal conjugates with respect to ABC .*

Before we begin, we'll let the tangency points of $\mathcal{E}$ with BC, CA, AB be D, E, F .

1. Reflect Q over BC, CA, AB to get Q_A, Q_B, Q_C . Prove that Q_A, D, P are collinear.
2. Prove that Q_A, Q_B, Q_C all lie on a circle ω centered at P .
3. Take ω to a circle ω' via a homothety with factor $1/2$ centered at Q . Show that ω' is the pedal circle of both P and Q , thus demonstrating that they are isogonal conjugates.

3.2 Exercises

1. Point O is inside $\triangle ABC$. The feet of perpendicular from O to BC, CA, AB are D, E, F . Perpendiculars from A and B respectively to EF and FD meet at P . Let H be the foot of perpendicular from P to AB . Prove that D, E, F, H are concyclic.
2. Let P be a given point inside quadrilateral $ABCD$. Points Q_1 and Q_2 are located within $ABCD$ such that

$$\angle Q_1BC = \angle ABP, \quad \angle Q_1CB = \angle DCP, \quad \angle Q_2AD = \angle BAP, \quad \angle Q_2DA = \angle CDP.$$

Prove that $\overline{Q_1Q_2} \parallel \overline{AB}$ if and only if $\overline{Q_1Q_2} \parallel \overline{CD}$.

3. Let ABC be a triangle, and let P and Q be a pair of isogonal conjugates with respect to ABC . Let ω_1 and ω_2 denote the circumcircles of triangles BPC and BQC , respectively. Suppose the circle with diameter $\overline{AP}$ intersects ω_1 again at M , and line AM intersects ω_1 again at X . Similarly, suppose the circle with diameter $\overline{AQ}$ intersects ω_2 again at N , and line AN intersects ω_2 again at Y . Prove that lines MN and XY are parallel.

4. Let O be the circumcenter and H the orthocenter of an acute triangle ABC . Show that there exist points D , E , and F on sides BC , CA , and AB respectively such that

$$OD + DH = OE + EH = OF + FH$$

and the lines AD , BE , and CF are concurrent.

5. Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. Let E be the midpoint of AC . Denote by ω and Ω the circumcircles of the triangles ABE and CDE , respectively. Let P be the crossing point of the tangent to ω at A with the tangent to Ω at D . Prove that PE is tangent to Ω .
6. Let ABC be an acute angled triangle satisfying the conditions $AB > BC$ and $AC > BC$. Denote by O and H the circumcentre and orthocentre, respectively, of the triangle ABC . Suppose that the circumcircle of the triangle AHC intersects the line AB at M different from A , and the circumcircle of the triangle AHB intersects the line AC at N different from A . Prove that the circumcentre of the triangle MNH lies on the line OH .
7. Consider a tetrahedron $ABCD$. There exists a sphere ω inside $ABCD$ tangent to all 4 of its faces, and there exists a sphere Ω outside $ABCD$, which is tangent to the plane made by ABC at the interior of ABC , and tangent to all three planes made by the other faces at points not in the interiors of the faces themselves. Suppose ω is tangent to ABC at P , and Ω is tangent to ABC at Q . Prove that P and Q are isogonal conjugates with respect to ABC .
8. A triangular pyramid $ABCD$ is given. A sphere ω_A is tangent to the face BCD and to the planes of other faces in points don't lying on faces. Similarly, sphere ω_B is tangent to the face ACD and to the planes of other faces in points don't lying on faces. Let K be the point where ω_A is tangent to ACD , and let L be the point where ω_B is tangent to BCD . The points X and Y are chosen on the prolongations of AK and BL over K and L such that $\angle CKD = \angle CXD + \angle CBD$ and $\angle CLD = \angle CYD + \angle CAD$. Prove that the distances from the points X , Y to the midpoint of CD are the same.

4 Isogonal Conjugates in Quadrilaterals

As a final topic, let's consider how isogonal conjugates work in quadrilaterals. As a start, verify yourself that arbitrary points do *not* necessarily have isogonal conjugates with respect to quadrilaterals. Here's a start on how to determine when points in quadrilaterals *do* have isogonal conjugates.

Theorem 4.1. *Consider a quadrilateral $ABCD$ and a point P . Let E, F, G, H be the projections of P to AB, BC, CD, DA , respectively. Then $EFGH$ is cyclic (i.e. P has a Pedal quadrilateral with respect to $ABCD$) iff*

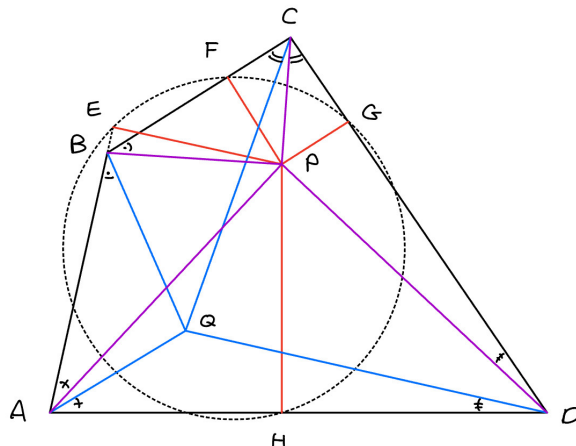
$$\angle APB + \angle CPD = 180^\circ.$$

To prove this, consider using angle chasing.

Using this, we can characterize when points have isogonal conjugates with respect to quadrilaterals.

Theorem 4.2. *Consider a quadrilateral $ABCD$ and a point P not on any of the lines $\{AB, AC, AD, BC, BD, CD\}$. Then P has an isogonal conjugate with respect to $ABCD$ iff*

$$\angle APB + \angle CPD = 180^\circ.$$



1. Let $E = AB \cap CD$. Show that P has an isogonal conjugate with respect to $ABCD$ iff the isogonal conjugates of P with respect to EAD and EBC coincide.
2. Show that latter condition is equivalent to P having a pedal circle with respect to $ABCD$.
3. Now finish.

This theorem is very nice on its own. But here's a contest problem demonstrating its usefulness!

Example 4.3. Let $ABCDEF$ be a hexagon inscribed in a circle Ω such that triangles ACE and BDF have the same orthocenter. Suppose that segments BD and DF intersect CE at X and Y , respectively. Show that there is a point common to Ω , the circumcircle of $DX Y$, and the line through A perpendicular to CE .

1. Identify the point in the question.
2. Let the common orthocenter be H , and let the center of Ω be O . Prove that it suffices to show that $\angle XHY = \angle BDF$.
3. Prove that H (and O) have pedal circles in $BXYF$, by considering certain *nine-point circles*.
4. Prove that $\angle XHY + \angle BHF = 180^\circ$, and finish.
5. Bonus question (interesting but unrelated to handout): how do you construct this diagram with straightedge and compass?

4.1 Exercises

1. In triangle ABC , let P and Q be two interior points such that $\angle ABP = \angle QBC$ and $\angle ACP = \angle QCB$. Point D lies on segment BC . Prove that $\angle APB + \angle DPC = 180^\circ$ if and only if $\angle AQC + \angle DQB = 180^\circ$.
2. There is given a convex quadrilateral $ABCD$. Prove that there exists a point P inside the quadrilateral such that

$$\angle PAB + \angle PDC = \angle PBC + \angle PAD = \angle PCD + \angle PBA = \angle PDA + \angle PCB = 90^\circ$$

if and only if the diagonals AC and BD are perpendicular.

3. Consider a quadrilateral $ABCD$. Prove that there exists a point X on segment $\overline{BD}$ such that $\angle BAC = \angle XAD$ and $\angle BCA = \angle XCD$ if and only if there exists a point Y on segment $\overline{AC}$ such that $\angle CBD = \angle YBA$ and $\angle CDB = \angle YDA$.

4. A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. Point X lies inside $ABCD$ so that

$$\angle XAB = \angle XCD \quad \text{and} \quad \angle XBC = \angle XDA.$$

Prove that $\angle BXA + \angle DXC = 180^\circ$.

5. Consider a triangle ABC with $AB + AC = 2AC$. Suppose H is its orthocenter, I its incenter, and M the midpoint of BC . Prove that $\angle HIM = 90^\circ$.
6. Let $ABCD$ be a convex quadrilateral whose sides AD and BC are not parallel. Suppose that the circles with diameters AB and CD meet at points E and F inside the quadrilateral. Let ω_E be the circle through the feet of the perpendiculars from E to the lines AB, BC and CD . Let ω_F be the circle through the feet of the perpendiculars from F to the lines CD, DA and AB . Prove that the midpoint of the segment EF lies on the line through the two intersections of ω_E and ω_F .

5 Solutions

Example 5.1. Consider a triangle ABC , and points $D, E \in BC$. Then (ABC) and (ADE) are tangent (at A) iff AD and AE are isogonal with respect to $\angle BAC$.

Proof. Let $AD \cap (ABC) = X, AE \cap (ABC) = Y$. Then AD and AE being isogonal is equivalent to arcs BX and CY subtending the same angle, which is equivalent to BC being parallel to XY . Now if BC is parallel to XY then the homothety from A taking ADE to AXY takes (ADE) to (ABC) , and so the two circles are tangent at A . Similarly the tangency condition implies that the homothety taking D to X also takes E to Y , so it's equivalent to BC and XY being parallel. $\square$

Example 5.2. Consider a triangle ABC , and points $D \in BC$ and $E \in (ABC)$. If AD and AE are isogonal with respect to $\angle BAC$, then ABD and AEC are *spirally similar*.

Proof. Observe that $\angle BAD = \angle CAE$ by definition, and $\angle ABD = \angle ABC = \angle AEC$, so the result follows. $\square$

Example 5.3. Consider a triangle ABC . Let M be the midpoint of BC , and let the tangents to (ABC) from B and C intersect at K . Then AK and AM are isogonal with respect to $\angle BAC$.

Proof. Let $AK \cap (ABC) = T$. Then

$$\frac{TB}{TC} = \frac{TB}{TK} \frac{TK}{TC} = \frac{AB}{BK} \frac{CK}{AC} = \frac{AB}{AC}.$$

Now suppose N is on BC such that AK and AN are isogonal with respect to $\angle BAC$. Then by the previous spiral similarity,

$$\frac{BN}{NC} = \frac{BN}{AN} \frac{AN}{CN} = \frac{CT}{AC} \frac{AB}{BT} = 1.$$

Thus $N = M$ as desired. $\square$

Example 5.4. Let $ABCD$ be a cyclic quadrilateral whose diagonals AC and BD meet at E . The extensions of the sides AD and BC beyond A and B meet at F . Let G be the point such that $ECGD$ is a parallelogram, and let H be the image of E under reflection in AD . Prove that D, H, F, G are concyclic.

Proof. By the isogonal line lemma, we have that $FCGD \sim FAEB$ (you may also prove this directly via angle chasing). Thus $\angle FGD = \angle FEB = 180^\circ - \angle FED = 180^\circ - \angle FHD$, and we are done. $\square$

Example 5.5. (Pascal's Theorem) Given six points A, B, C, D, E, F on a circle, show that the three points $X = AB \cap DE, Y = BC \cap EF, Z = CD \cap FA$ are collinear.

Proof. Observe that XDA and XBE are similar. Construct Z' such that $XDAZ'$ and $XBEZ'$ are similar. Then

$$\begin{aligned}\angle Z'EB &= \angle ZAD = \angle FAD = \angle FED = \angle YEX \\ \angle Z'BE &= \angle ZDA = \angle CDA = \angle CBA = \angle YBX\end{aligned}$$

so Y and Z' are isogonal conjugates with respect to XBE . This gives us that

$$\angle BXY = \angle Z'XE = \angle ZXA,$$

so X, Y, Z are collinear, as desired. $\square$

Theorem 5.6. Consider a triangle ABC , as well as points P, Q which are isogonal conjugates with respect to ABC . Then P and Q share the same pedal circle.

Define $D_P, D_Q, E_P, E_Q, F_P, F_Q$ as in the walkthrough as the feet of the altitudes from P, Q to the sides of the triangle. Then

$$\angle AF_P E_P = \angle A P E_P = 90^\circ - \angle PAC = 90^\circ - \angle QAB = \angle A Q F_Q = \angle A E_Q F_Q,$$

so $E_P E_Q F_P F_Q$ is cyclic. Similarly $D_P D_Q E_P E_Q, D_P D_Q F_P F_Q$ are cyclic. If these three circumcircles are distinct, then by the Radical Axis theorem, their radical axes concur. But their radical axes are AB, AC , and BC , so since they do not concur we find that $D_P D_Q E_P E_Q F_P F_Q$ is cyclic, so P, Q share a pedal circle, as desired.

Theorem 5.7. *Consider a triangle ABC , and an ellipse $\mathcal{E}$ inscribed within it. Then the foci P and Q of $\mathcal{E}$ are isogonal conjugates with respect to ABC .*

Proof. Reflect Q over BC, CA, AB to get Q_A, Q_B, Q_C . Also let $\mathcal{E}$ touch BC, CA, AB at D, E, F . Then we have $\angle PDB = \angle QDC = \angle Q_A DC$, so P, D, Q_A are collinear (and similarly for Q_B, Q_C). Now $PD + DQ = PE + EQ + PF + FQ = x$ for some fixed x , but $PD + DQ = PD + DQ_A = PQ_A$. Thus $PQ_A = PQ_B = PQ_C = x$, and so there exists a circle ω centered at P with radius x . Take a $1/2$ homothety about Q to send ω to ω' , and let O be the midpoint of PQ . Then ω' is centered at O and passes through the projections of Q to the sides of ABC . But the distances from O to the projections of P and Q to BC are the same; thus ω' also passes through the projections of P to the sides of ABC . Hence by the converse of Theorem 3.5 (which one may prove readily with angle chasing), we find that P and Q are isogonal conjugates, as desired. $\square$

Theorem 5.8. *Consider a quadrilateral $ABCD$ and a point P not on any of the lines $\{AB, AC, AD, BC, BD, CD\}$. Then P has an isogonal conjugate with respect to $ABCD$ iff*

$$\angle APB + \angle CPD = 180^\circ.$$

Proof. Let $E = AD \cap BC$. By angle chasing, one may show that the projections of P to the sides of $ABCD$ are cyclic iff $\angle APB + \angle CPD = 180^\circ$. This former condition is equivalent to P having the same pedal circles with respect to EAB and ECD . This latter condition is equivalent to the isogonal conjugates of P with respect to EAB and ECD being the same point, which is equivalent to P having an isogonal conjugate with respect to $ABCD$. Hence we are done. $\square$

Example 5.9. Let $ABCDEF$ be a hexagon inscribed in a circle Ω such that triangles ACE and BDF have the same orthocenter. Suppose that segments BD and DF intersect CE at X and Y , respectively. Show that there is a point common to Ω , the circumcircle of $DEXY$, and the line through A perpendicular to CE .

Proof. Let H be the shared orthocenter, O the center of Ω , and N the midpoint of OH . Then the reflection A' of H across CE lies on the line through A perpendicular to CE as well as Ω , so it suffices to show $\angle XA'Y = \angle XDY$. But $\angle XA'Y = \angle XHY$ and $\angle XDY = 180^\circ - \angle BHF$, so it suffices to show that H has an isogonal conjugate with respect to $BXYF$. Now the pedal circle of H with respect to ACE and BDF are their respective nine-point circles. These circles are both centered at N and have radius half of that of Ω , so they coincide. Hence H has a pedal circle with respect to $BXYF$, which by Theorem 4.1 is equivalent to H having an isogonal conjugate with respect to $BXYF$, as desired. $\square$